A Multi-Layer Structural Framework for Bitcoin Treasury Vehicles: Implied Accretion, Leverage Elasticity, and Capital Stack Dynamics

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Abstract

We develop a multi-layer stochastic structural model for valuing publicly traded companies whose primary asset is Bitcoin and that employ a leverage flywheel to grow per-share BTC exposure. The canonical example is Strategy (formerly MicroStrategy, MSTR), which as of late 2025 holds over 649,000 BTC financed through a complex capital stack of common equity, convertible bonds, and multiple series of perpetual preferred stock. No formal structural framework exists for these "Bitcoin treasury vehicles," leaving investors without rigorous tools to assess whether equity premia reflect achievable BTC accretion or unsustainable leverage.

Our model couples four stochastic processes under the physical measure: (i) BTC price dynamics with jump-diffusion, (ii) an Ornstein–Uhlenbeck process for the equity-over-NAV log-premium capturing market sentiment and mean-reversion, (iii) a compound Poisson process for BTC holding accumulation driven by the leverage flywheel, and (iv) share-count dynamics from equity issuance and convertible conversion. We derive five novel diagnostic indicators—the Implied BTC Growth Rate (IBGR), Implied Leverage Elasticity (ILE), Total Equity Elasticity (TEE), Premium Mean-Reversion Index (PMRI), and Implied Funding Requirement Distribution (IFRD)—that decompose valuation into structural balance-sheet leverage, sentiment premium, and accretion efficiency.

Calibrating to Strategy's publicly available data through December 2025, we find: the model implies an 18.3% annualized BTC accretion rate (18.4% per-share), a structural leverage elasticity of 1.17, and a three-year survival probability exceeding 99.4%. The current premium sits 1.5 standard deviations below its long-run mean, suggesting mean-reversion potential. We extend the framework conceptually to accommodate perpetual preferred stock layers (STRK, STRF) via a priority waterfall, laying groundwork for a complete multi-tranche capital structure analysis.

1 Introduction

1.1 Motivation

Strategy (formerly MicroStrategy, ticker MSTR) has become the paradigmatic "Bitcoin treasury vehicle"—a publicly traded company whose equity value derives overwhelmingly from its Bitcoin holdings rather than operating cash flows. Beginning with an initial 21,454 BTC purchase in August 2020, the company has accumulated over 649,000 BTC by late 2025, financed through a distinctive flywheel of at-the-market (ATM) equity offerings, convertible bond issuances, and most recently, multiple series of perpetual preferred stock. The resulting capital structure—spanning common equity, convertible debt, and preferred tranches including STRK (8% convertible preferred), STRF (10% non-convertible preferred), STRC, STRD, and STRE (EUR-denominated)—has no close precedent in corporate finance.

Strategy's equity has persistently traded at a premium to its net asset value (NAV), with this premium fluctuating dramatically over time. The premium reflects the market's assessment of management's ability to continue acquiring BTC accretively—that is, growing BTC per share even as new shares are issued—combined with the embedded leverage from the debt stack and the optionality of the premium itself. Understanding whether a given premium level is "justified" by achievable accretion rates, or whether it reflects unsustainable speculation, is the central question facing investors in MSTR and the growing number of imitators adopting similar strategies.

Despite the significance of this question, no formal structural model exists for Bitcoin treasury vehicles. Traditional corporate structural models in the Merton (1974) tradition treat the firm's assets as following geometric Brownian motion with a fixed capital structure, but Strategy's assets are a single volatile cryptocurrency whose holdings *grow endogenously* in response to the very premium the model seeks to explain. The leverage flywheel creates a feedback loop: a high premium enables equity issuance, which funds BTC purchases, which (if accretive) supports the premium. This circularity demands a purpose-built framework.

1.2 Research Gap

The existing literature offers partial tools but no integrated model. Merton's (1974) structural credit model and its extensions (Black and Cox, 1976; Leland, 1994) provide the foundation for modeling equity as a call option on firm assets, but assume exogenous asset dynamics and static capital structures. The growing literature on Bitcoin valuation (Biais et al., 2023; Cong et al., 2021) addresses cryptocurrency pricing but not the corporate wrapper. Convertible bond pricing models (Tsiveriotis and Fernandes, 1998) handle individual instruments but not the full multi-layer stack with endogenous asset accumulation.

The closest practical analogs—closed-end funds trading at premia or discounts to NAV—have been studied extensively (Lee et al., 1991; Pontiff, 1996), but these funds have static portfolios and do not employ leverage flywheels. Strategy's defining feature is that its portfolio is *not* static: the premium drives accumulation, and accumulation supports the premium. This endogeneity is the core modeling challenge we address.

1.3 Contributions

This paper makes the following contributions:

1. **A stochastic structural framework for BTC treasury vehicles.** We develop a multi-process model under the physical measure that jointly evolves BTC price, the equity-over-NAV premium, BTC holdings, and share count. The premium follows an Ornstein–Uhlenbeck process capturing mean-reversion, while holdings evolve via a compound Poisson process linked to the flywheel. The model nests several special cases including static treasury, pure-equity financing, and standard Merton-style structural models.

2. **Five novel diagnostic indicators.** We derive the Implied BTC Growth Rate (IBGR), which measures the model-implied annualized rate of BTC accumulation both in total and per-share; the Implied Leverage Elasticity (ILE), which decomposes equity sensitivity to BTC into a pure balance-sheet component; the Total Equity Elasticity (TEE), which adds the premium's response to BTC; the Premium Mean-Reversion Index (PMRI), a z-score measuring how far the current premium deviates from its long-run level; and the Implied Funding Requirement Distribution (IFRD), the probability distribution of capital needed to sustain the implied accretion path.

3. **Empirical calibration to Strategy.** We calibrate the full model to publicly available data through December 2025, estimating all structural parameters and computing each indicator. The calibration reveals economically interpretable dynamics: an 18.3% annualized BTC accretion rate, leverage elasticity of 1.17 reflecting moderate balance-sheet leverage, and a premium currently 1.5 stationary standard deviations below its long-run mean. The three-year survival probability exceeds 99.4%, suggesting the current capital structure is robust to stress.

4. **Extension to multi-layer preferred capital stacks.** We outline how the framework extends to accommodate preferred stock layers through a priority waterfall that respects perpetual dividend obligations and conversion optionality, establishing a foundation for full multi-tranche valuation.

1.4 Paper Outline

The remainder of the paper is organized as follows. Section 3 presents the structural model: BTC price dynamics, the premium process, holding dynamics, debt and share-count modeling, and the derivation of all five indicators. Section 5 reports the calibrated parameters and computed indicators with economic interpretation. Section 6 discusses implications for investors, model limitations, and extensions. Section 7 concludes.

2 Literature Review

This paper draws on several distinct strands of the finance literature. We synthesize insights from structural credit models, closed-end fund pricing, convertible bond theory, cryptocurrency

finance, and capital structure research to build a framework suited to the unique characteristics of Strategy Inc (formerly MicroStrategy) as a BTC-backed equity vehicle.

2.1 Structural Credit Models

The foundational framework for linking firm asset values to security prices originates with Merton (1974), who models equity as a European call option on the firm's assets with a strike equal to the face value of debt. This insight—that limited liability creates option-like payoffs for equity holders—is central to our model's treatment of NAV with a limited liability floor: $\text{NAV}_t = (H_t S_t - D_t)^+$.

? extend Merton's framework to incorporate safety covenants and subordination, both relevant to Strategy's multi-layered capital structure with preferred stock tranches of differing seniority. **?** and **?** develop models with endogenous bankruptcy and optimal capital structure, introducing the trade-off between tax shields and distress costs that informs our survival probability analysis. More recently, **?** provide empirical evidence on the performance of structural models, finding that they substantially underpredict credit spreads—the so-called "credit spread puzzle"—suggesting that factors beyond pure leverage drive credit risk.

In our setting, the single-asset nature of Strategy's balance sheet (essentially only BTC) makes structural models particularly apt: the firm's asset volatility is directly observable via Bitcoin price dynamics, unlike diversified corporations where asset volatility must be inferred. However, the presence of multiple layers of preferred stock with distinct priority claims extends the standard two-class (debt/equity) Merton framework into a multi-tranche setting analogous to structured credit.

2.2 NAV Premium and Discount Dynamics

Strategy trades at a persistent premium to its Bitcoin net asset value, a phenomenon with close parallels to the closed-end fund (CEF) pricing puzzle. **?** propose that investor sentiment drives CEF discounts and premia, with noise trader risk preventing arbitrage from eliminating mispricings. **?** documents that CEF premia predict future returns, consistent with mean-reversion—the empirical pattern we capture through our Ornstein–Uhlenbeck premium process π_t.

? offer a liquidity-based explanation: CEFs provide access to illiquid assets, justifying a premium when the underlying is hard to access directly. This theory has particular relevance to Strategy, which offers leveraged, operationally-managed BTC exposure that individual investors may find difficult or costly to replicate, especially pre-2024 when spot Bitcoin ETFs were unavailable in the United States. **?** provide a comprehensive survey of CEF pricing anomalies, noting that premia tend to be larger for funds holding exotic or hard-to-value assets—a characterization that arguably applies to Bitcoin.

The introduction of spot Bitcoin ETFs in January 2024 (**?**) represents a structural break in the accessibility of BTC exposure, potentially compressing Strategy's premium over time. Our mean-reverting premium model with regime-dependent long-run mean θ can accommodate such shifts. The premium dynamics we observe—with π_t negatively correlated with BTC returns

($\rho < 0$), reflecting premium compression on BTC rallies—extend the CEF literature's sentiment patterns to the leveraged treasury setting.

2.3 Bitcoin and Cryptocurrency Finance

The academic literature on Bitcoin pricing has grown rapidly since **?**. **?** establish that cryptocurrency returns are driven by network and momentum factors distinct from traditional asset pricing factors, implying that BTC exposure introduces novel risk dimensions to Strategy's equity. **?** document arbitrage frictions in cryptocurrency markets that can amplify volatility, relevant to our BTC price dynamics specification.

On volatility modeling, **?** compares GARCH specifications for Bitcoin and finds that models with both short- and long-run volatility components best capture BTC dynamics. **?** provide a general equilibrium model of Bitcoin pricing that links fundamentals (mining costs, network adoption) to price, offering theoretical grounding for the drift parameter μ_S in our $\mathbb{P}$-measure BTC dynamics. The jump-diffusion specification we employ for BTC prices under $\mathbb{Q}$ follows **?** and **?**, accommodating the well-documented fat tails and discontinuous price movements in cryptocurrency markets (**?**).

2.4 Bitcoin as a Corporate Treasury Asset

Strategy's pioneering adoption of Bitcoin as a treasury reserve asset, beginning in August 2020, has spawned a small but growing literature. **?** study MicroStrategy's treasury strategy and its effects on firm valuation, documenting the reflexive relationship between BTC acquisitions, equity premium, and subsequent capital raises that we formalize as the "flywheel" mechanism. **?** examine more broadly whether corporate cryptocurrency holdings affect firm value, finding heterogeneous effects depending on the scale of holdings relative to firm size.

Our model's BTC holding dynamics—with accumulation rate α driven by the premium π_t^+ and discrete financing jumps—formalizes the strategic feedback loop whereby a high premium enables accretive capital raises, which increase BTC per share, which supports the premium. This is a novel contribution: while practitioner analyses have described this flywheel qualitatively, we provide the first stochastic model of its dynamics suitable for Monte Carlo simulation and quantitative indicator extraction.

2.5 Convertible Bond Pricing and Hybrid Securities

Strategy's capital structure includes approximately \$8.2 billion in convertible notes alongside five series of preferred stock with varying terms. The convertible bond pricing literature, beginning with **?** and **?**, provides the theoretical basis for valuing these hybrid securities. **?** extend the framework to incorporate credit risk, decomposing convertible value into debt and equity components—an approach we use to define "effective debt" D_t^{eff} that adjusts for deep-in-the-money convertibles. **?** provide empirical evidence that convertible bonds are systematically underpriced, a finding with implications for the true leverage embedded in Strategy's capital structure.

The preferred stock layer adds further complexity. **?** document that preferred stocks exhibit hybrid risk characteristics, with returns driven by both interest rate and equity market factors. Strategy's preferred tranches span from pure fixed-income (STRF, STRD) to equity-linked (STRK with conversion into MSTR common), with the variable-rate STRC representing an innovative instrument that adjusts dividends to stabilize price near par—a mechanism without close precedent in the academic literature.

2.6 Capital Structure and Leverage

The classical capital structure theories of **?** and **?** provide the backdrop against which Strategy's aggressive leveraging of a single volatile asset must be understood. **?** document that firms time equity issuance to exploit high valuations—precisely the behavior we model through premium-dependent financing intensity $\lambda_M(\pi_t)$.

Strategy's capital structure is unusual in several respects that extend standard theory: (i) the firm's assets are a single, publicly-traded, highly volatile cryptocurrency; (ii) leverage is deliberately and systematically increased to accumulate more of the same asset; (iii) the financing flywheel creates a reflexive feedback loop between equity valuation and asset accumulation; and (iv) the multi-layered preferred stock structure creates a priority waterfall with complex interactions between dividend obligations, conversion optionality, and residual equity claims.

These features motivate our multi-tranche extension of the Merton framework with endogenous asset accumulation, a contribution at the intersection of structural credit modeling and corporate treasury management.

2.7 Simulation Methods

Our quantitative indicators—including the Implied Funding Requirement Distribution (IFRD), per-share Implied BTC Growth Rate (IBGR), and survival probabilities—require Monte Carlo simulation of correlated stochastic processes. We follow the methods described in **?**, employing Cholesky decomposition for correlated Brownian motions and thinning algorithms for state-dependent Poisson jump processes. The Ornstein–Uhlenbeck process for the log-premium follows the exact discretization of **?**, ensuring that our discrete-time simulation preserves the known transition density of the continuous-time process.

3 Methodology

This section presents the structural model in full. We begin with the probability space and state variables, then specify each stochastic process, derive the equity valuation identity, and finally define the five diagnostic indicators.

3.1 Setup: Probability Space and State Variables

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0})$ equipped with two measures:

- $\mathbb{P}$: the physical (historical) measure, used for premium dynamics, management behavior (holdings accumulation), survival analysis, and all indicator computations.

- $\mathbb{Q}$: the risk-neutral measure, available for calibrating BTC volatility and jump parameters to options data if desired.

The model's state variables are:

- S_t: BTC spot price (BTC-USD).
- H_t: Strategy's BTC holdings (in BTC).
- D_t: outstanding debt (face value).
- N_t^{sh}: shares of common equity outstanding.
- π_t: log-premium of equity over NAV.

From these primitives, we derive:

$$A_t = H_t S_t, \qquad \text{(BTC asset value)} \tag{1}$$

$$\text{NAV}_t = (A_t - D_t)^+ = \max(H_t S_t - D_t, 0), \qquad \text{(net asset value)} \tag{2}$$

$$E_t = P_t N_t^{\text{sh}}, \qquad \text{(equity market capitalization)} \tag{3}$$

$$B_t = H_t / N_t^{\text{sh}}, \qquad \text{(BTC per share)} \tag{4}$$

where P_t is the share price.

To avoid log singularities when computing the historical premium from data, we introduce a floor $\varepsilon > 0$:

$$\text{NAV}_t^{\text{clip}} = \max(A_t - D_t, \varepsilon), \tag{5}$$

and define the log-premium as:

$$\pi_t = \log\left(\frac{E_t}{\text{NAV}_t^{\text{clip}}}\right). \tag{6}$$

Periods where $A_t \approx D_t$ are flagged as a "distress regime" and handled separately in calibration.

3.2 BTC Price Dynamics

Under the risk-neutral measure $\mathbb{Q}$, for option calibration we consider a jump-diffusion:

$$\frac{dS_t}{S_{t^-}} = (r_t - q_S - \lambda_S k_S)\, dt + \sigma_S(S_t, t)\, dW_t^{S,\mathbb{Q}} + \left(e^{Y_S} - 1\right) dN_t^S, \tag{7}$$

where r_t is the risk-free rate, q_S is any BTC carry term (set to zero at baseline), $W_t^{S,\mathbb{Q}}$ is a standard Brownian motion under $\mathbb{Q}$, N_t^S is a Poisson process with intensity λ_S, Y_S is the log-jump size with distribution ν_S, $k_S = \mathbb{E}[e^{Y_S} - 1]$ is the jump compensator, and $\sigma_S(S_t, t)$ is a potentially state-dependent volatility.

For the physical-measure simulations that drive our indicator computations, we use a simpler geometric Brownian motion calibrated to historical BTC returns:

$$\frac{dS_t}{S_t} = \mu_S\, dt + \sigma_S\, dW_t^{S,\mathbb{P}}, \tag{8}$$

with constant drift μ_S and volatility σ_S estimated from daily log-returns.

3.3 Premium Dynamics

The log-premium π_t is modeled as an Ornstein–Uhlenbeck (OU) process under the physical measure, capturing the empirical observation that premia to NAV exhibit mean-reversion:

$$d\pi_t = \kappa(\theta - \pi_t)\,dt + \sigma_\pi\,dW_t^{\pi,\mathbb{P}} + \eta_\pi\,dJ_t^\pi, \tag{9}$$

where $\kappa > 0$ is the mean-reversion speed, θ is the long-run mean log-premium, σ_π is the diffusive volatility, $W_t^{\pi,\mathbb{P}}$ is a Brownian motion under $\mathbb{P}$, and J_t^π is an optional pure-jump process with intensity λ_π (set to zero at baseline).

The BTC and premium Brownian motions are correlated:

$$\mathrm{corr}(dW_t^{S,\mathbb{P}}, dW_t^{\pi,\mathbb{P}}) = \rho. \tag{10}$$

This correlation is economically meaningful: a negative ρ indicates that the premium tends to compress when BTC rises (and vice versa), reflecting a "buy the rumor, sell the news" dynamic in NAV-premium relationships.

The exact discrete-time transition for daily step Δt is:

$$\pi_{t+\Delta t} = \theta + (\pi_t - \theta)e^{-\kappa\Delta t} + \sigma_\pi\sqrt{\frac{1-e^{-2\kappa\Delta t}}{2\kappa}}\,Z_\pi, \tag{11}$$

where (Z_S, Z_π) are bivariate standard normal with $\mathrm{corr}(Z_S, Z_\pi) = \rho$.

3.4 BTC Holding Dynamics: The Leverage Flywheel

The defining feature of a Bitcoin treasury vehicle is that holdings are *endogenous*: management acquires BTC using proceeds from equity and debt issuance, with the pace of acquisition linked to the premium. We model this via:

$$dH_t = \alpha\,\pi_t^+\,dt + dH_t^{\text{jump}}, \qquad \pi_t^+ = \max(\pi_t, 0), \tag{12}$$

where the continuous drift captures routine accumulation scaled by the positive premium, and the jump component captures discrete financing events:

$$dH_t^{\text{jump}} = \sum_k \Delta H_{\tau_k}\,\delta_{t=\tau_k}, \qquad \Delta H_{\tau_k} \geq 0. \tag{13}$$

Here $\{\tau_k\}$ are event times of a Poisson process M_t with intensity λ_M, and ΔH_{τ_k} are BTC amounts purchased. The jump sizes are drawn from an empirical distribution $\mathcal{D}_H$ calibrated to observed purchase events.

In discrete time:

$$H_{t+\Delta t} = H_t + \alpha\pi_t^+\Delta t + \sum_{i=1}^{N_t^M}\Delta H_{t,i}, \qquad N_t^M \sim \text{Poisson}(\lambda_M\Delta t). \tag{14}$$

When the data suggest $\alpha \approx 0$ (as in our calibration), the model reduces to a pure compound

Poisson accumulation process, with all BTC growth occurring through discrete financing events. The intensity or size distribution can be further enriched to depend on leverage or premium levels.

3.5 Debt and Share-Count Dynamics

Debt. At baseline, total outstanding debt is modeled as deterministic and piecewise-constant:

$$D_t = \sum_{j=1}^{J} D_j \mathbf{1}_{\{t<T_j\}}, \tag{15}$$

where each debt instrument j has face value D_j and maturity T_j. This conservative treatment ignores the equity-like behavior of deep in-the-money convertibles, providing a solvency lower bound.

Share count. Shares outstanding evolve through equity issuance and convertible conversion:

$$dN_t^{\text{sh}} = dN_t^{(\text{equity})} + dN_t^{(\text{conv})}, \tag{16}$$

where both components are jump processes aligned with financing and conversion events. BTC per share is then:

$$B_t = \frac{H_t}{N_t^{\text{sh}}}. \tag{17}$$

The per-share metric B_t is central to assessing whether the flywheel is *accretive*: if B_t grows over time despite share dilution, each share commands more BTC.

3.6 Equity Valuation Identity

The structural relation linking equity, assets, debt, and premium is:

$$E_t = e^{\pi_t} \cdot (H_t S_t - D_t)^+. \tag{18}$$

This identity holds exactly when $\text{NAV}_t > \varepsilon$ (i.e., outside the distress regime where $\text{NAV}_t^{\text{clip}} = \text{NAV}_t$). It decomposes equity market capitalization into NAV (the intrinsic balance-sheet value) and the exponential of the log-premium (the market's sentiment multiplier). The premium absorbs all factors not captured by current-snapshot NAV: expected future accretion, management optionality, liquidity, and speculative demand.

3.7 Diagnostic Indicators

We derive five indicators that together characterize the structural health and valuation of a Bitcoin treasury vehicle.

3.7.1 Implied BTC Growth Rate (IBGR)

The total IBGR measures the expected annualized rate of BTC accumulation under the physical measure:

$$\mu_H = \frac{1}{H_0}\mathbb{E}^{\mathbb{P}}\left[\frac{H_{\Delta t} - H_0}{\Delta t} \,\middle|\, S_0, \pi_0, H_0\right] \approx \frac{1}{H_0}\left(\alpha\pi_0^+ + \lambda_M \mathbb{E}[\Delta H]\right). \tag{19}$$

The per-share IBGR accounts for dilution:

$$\mu_{H/N} = \frac{1}{B_0}\mathbb{E}^{\mathbb{P}}\left[\frac{B_{\Delta t} - B_0}{\Delta t} \,\middle|\, S_0, \pi_0, H_0, N_0^{\text{sh}}\right], \tag{20}$$

estimated via Monte Carlo simulation of joint (H_t, N_t^{sh}) paths. Positive $\mu_{H/N}$ indicates accretive growth—the flywheel is delivering more BTC per share despite dilution.

3.7.2 Implied Leverage Elasticity (ILE)

The ILE captures the pure balance-sheet sensitivity of equity to BTC price, holding the premium fixed:

$$\text{ILE}_t = \beta_{LS}(t) = \left.\frac{\partial \log E_t}{\partial \log S_t}\right|_{\pi_t \text{ fixed}} = \frac{H_t S_t}{H_t S_t - D_t}, \tag{21}$$

for $H_t S_t > D_t$. An ILE of 1.17 means a 1% BTC move produces a 1.17% equity move from leverage alone, before any premium response.

3.7.3 Total Equity Elasticity (TEE)

The TEE incorporates the premium's empirical response to BTC:

$$\beta_{\text{TOT}}(t) = \frac{d \log E_t}{d \log S_t} = \gamma_{\pi S} + \beta_{LS}(t), \tag{22}$$

where $\gamma_{\pi S}$ is the regression coefficient of premium changes on BTC log-returns:

$$\gamma_{\pi S} \approx \frac{\text{Cov}(\Delta\pi_t, \Delta \log S_t)}{\text{Var}(\Delta \log S_t)}. \tag{23}$$

The TEE is what traders experience in their P&L. When $\gamma_{\pi S} < 0$, premium compression partially offsets leverage, and the TEE may be less than the ILE.

3.7.4 Premium Mean-Reversion Index (PMRI)

The PMRI standardizes the current premium against its OU stationary distribution:

$$\text{PMRI} = \frac{\pi_0 - \theta}{\sigma_\pi/\sqrt{2\kappa}}, \tag{24}$$

where $\sigma_\pi^2/(2\kappa)$ is the stationary variance of the OU process. Large positive values signal exuberance; large negative values signal distress or undervaluation.

3.7.5 Implied Funding Requirement Distribution (IFRD)

The IFRD characterizes the probability distribution of capital required to sustain the model-implied BTC accumulation trajectory over a horizon Δt:

$$G_{\Delta t} = \int_0^{\Delta t} S_u \, dH_u = \int_0^{\Delta t} S_u \alpha \pi_u^+ \, du + \sum_{0<u\leq\Delta t} S_u \Delta H_u. \tag{25}$$

In discrete simulation:

$$G_{\Delta t}^{(\text{path})} \approx \sum_{k=0}^{n-1} S_{t_k}(H_{t_{k+1}} - H_{t_k}). \tag{26}$$

Monte Carlo over simulated paths provides the empirical distribution of $G_{\Delta t}$, from which we extract the mean, standard deviation, and tail measures (e.g., $\text{VaR}_{95\%}$). Large IFRD values relative to the firm's financing capacity signal potential flywheel unsustainability.

3.8 Survival and Distress Analysis

Define a distress event as the first time BTC asset value breaches the debt boundary with buffer $\varepsilon_d \geq 0$:

$$\tau = \inf\{u \in [0, T] : H_u S_u \leq (1 + \varepsilon_d) D_u\}. \tag{27}$$

The survival probability over horizon T is:

$$\mathbb{P}(\tau > T) \approx \frac{\#\{\text{paths with no breach before } T\}}{\#\{\text{total simulated paths}\}}. \tag{28}$$

This Monte Carlo estimate accounts for the full joint dynamics of BTC price, premium, and holdings accumulation, providing a structurally informed solvency assessment that incorporates the flywheel's stabilizing (or destabilizing) effects.

3.9 Calibration Strategy

Premium OU parameters. Using the exact OU transition density, we estimate $(\kappa, \theta, \sigma_\pi)$ via maximum likelihood on the daily log-premium series, excluding distress-regime observations. The correlation ρ and premium-BTC sensitivity $\gamma_{\pi S}$ are estimated from empirical correlations and OLS regression of premium changes on BTC log-returns.

Holdings dynamics. We construct the BTC holding change series from public disclosures, identify financing event dates, and estimate jump intensity λ_M as the annualized event frequency and mean jump size $\mathbb{E}[\Delta H]$ from the empirical distribution of BTC purchased per event. The continuous drift α is estimated by regressing non-event holding changes on the positive premium; when statistically insignificant (as in our sample), we set $\alpha = 0$.

BTC price parameters. The drift μ_S and volatility σ_S are estimated from historical daily BTC log-returns under the physical measure.

4 Extended Structural Model: Capital Stack and Preferred Equity

We extend the single-debt baseline of Section 3 to accommodate Strategy's multi-layered capital structure, which as of early 2026 includes senior debt, convertible bonds, multiple series of perpetual preferred stock (both convertible and non-convertible), and common equity. The extensions below preserve all state variables and dynamics from the base model while enriching the liability side and its implications for leverage, survival, and BTC accretion.

4.1 Capital Stack and Priority Waterfall

Definition 4.1 (Capital Stack)**.** *At time t the firm's capital structure comprises:*

1. ***Senior debt*** $D_s(t)$*: term loans, secured notes, and operating liabilities with contractual priority.*
2. ***Convertible debt*** $D_c(t)$*: unsecured convertible notes with face values, coupons, maturities, and equity conversion rights.*
3. ***Non-convertible perpetual preferred stock*** $P_{nc}(t)$*: par value of outstanding non-convertible preferred (e.g. STRF at 10% cumulative dividend).*
4. ***Convertible perpetual preferred stock*** $P_c(t)$*: par value of outstanding convertible preferred (e.g. STRK at 8% cumulative dividend, convertible to 0.1 MSTR per share).*
5. ***Common equity*** $E(t)$*: residual claim.*

Total liabilities senior to common equity are

$$L_t = D_s(t) + D_c(t) + P_{nc}(t) + P_c(t). \tag{29}$$

Definition 4.2 (Liquidation Waterfall)**.** *In a hypothetical liquidation at time t with asset value $A_t = H_t S_t$, claims are satisfied in strict priority:*

$$\underbrace{D_s}_{\textit{senior debt}} \succ \underbrace{D_c}_{\textit{convertible debt}} \succ \underbrace{P_{nc}}_{\textit{non-conv. pfd.}} \succ \underbrace{P_c}_{\textit{conv. pfd.}} \succ \underbrace{E}_{\textit{common}}. \tag{30}$$

Each layer ℓ receives $\min(A_t^{(\ell)}, C_\ell)$, where C_ℓ is the face claim and $A_t^{(\ell)}$ is the residual asset value available after all senior layers are paid:

$$A_t^{(1)} = A_t, \qquad R_s(t) = \min\big(A_t^{(1)}, D_s(t)\big), \tag{31}$$

$$A_t^{(2)} = \big(A_t^{(1)} - D_s(t)\big)^+, \qquad R_{dc}(t) = \min\big(A_t^{(2)}, D_c(t)\big), \tag{32}$$

$$A_t^{(3)} = \big(A_t^{(2)} - D_c(t)\big)^+, \qquad R_{nc}(t) = \min\big(A_t^{(3)}, P_{nc}(t)\big), \tag{33}$$

$$A_t^{(4)} = \big(A_t^{(3)} - P_{nc}(t)\big)^+, \qquad R_c(t) = \min\big(A_t^{(4)}, P_c(t)\big), \tag{34}$$

$$A_t^{(5)} = \big(A_t^{(4)} - P_c(t)\big)^+, \qquad E_t^{res} = A_t^{(5)}. \tag{35}$$

The base-model NAV generalizes to

$$\mathrm{NAV}_t = \big(A_t - L_t\big)^+ = \big(H_t S_t - D_s(t) - D_c(t) - P_{nc}(t) - P_c(t)\big)^+, \tag{36}$$

and the log-premium identity becomes

$$\pi_t = \log\left(\frac{E_t}{\mathrm{NAV}_t^{\mathrm{clip}}}\right), \qquad \mathrm{NAV}_t^{\mathrm{clip}} = \max\big(A_t - L_t,\, \varepsilon\big). \tag{37}$$

Remark 4.3. *For periods where some convertible debt is deep in-the-money, one may define an* effective liability $L_t^{\mathrm{eff}} = D_s(t) + D_c^{\mathrm{eff}}(t) + P_{nc}(t) + P_c(t)$*, where* $D_c^{\mathrm{eff}}(t)$ *discounts the equity-like component (cf. the debt process in Section 3). This keeps the waterfall well-ordered while avoiding double-counting of the conversion value.*

4.2 Perpetual Preferred Stock Dynamics

Strategy has issued multiple series of perpetual preferred stock with fixed cumulative dividends. Unlike debt, preferred stock has no maturity; however, failure to pay dividends triggers cumulative arrearages and potentially accelerated conversion or governance consequences.

Definition 4.4 (Preferred Dividend Obligations)**.** *Let* $\mathcal{P} = \{p_1, \ldots, p_M\}$ *denote the* M *outstanding preferred series, each with par value* F_{p_i}*, number of shares* n_{p_i}*, and contractual annual dividend rate* δ_{p_i}*. The aggregate annual preferred dividend obligation is*

$$\Phi_t = \sum_{i=1}^{M} \delta_{p_i}\, F_{p_i}\, n_{p_i}(t). \tag{38}$$

For the current capital structure, the dominant terms are:

- *STRK:* $\delta_{\mathrm{STRK}} = 0.08$*, convertible.*
- *STRF:* $\delta_{\mathrm{STRF}} = 0.10$*, non-convertible.*

Definition 4.5 (Dividend Coverage Ratio)**.** *The* Dividend Coverage Ratio *at time* t *measures the firm's capacity to service preferred dividends from BTC-related cash flows:*

$$\mathrm{DCR}_t = \frac{\Delta A_t^+ + \mathrm{CF}_t^{\mathrm{ops}}}{\Phi_t\, \Delta t}, \tag{39}$$

where $\Delta A_t^+ = \max(\Delta A_t,\, 0)$ *captures realized BTC appreciation over the period* $[t, t + \Delta t]$ *and* $\mathrm{CF}_t^{\mathrm{ops}}$ *is operating cash flow from the legacy software business. The firm is in* dividend stress *when* $\mathrm{DCR}_t < 1$*.*

Proposition 4.6 (Extended Distress Threshold)**.** *Under the extended capital stack, the distress threshold (replacing the base-model condition* $A_t \le (1+\varepsilon)D_t$*) incorporates both insolvency and dividend-default risk:*

$$A_t \;<\; D_s(t) + D_c(t) + \mathrm{PV}_t\big(\textit{preferred dividends}\big) + \varepsilon\big(P_{nc}(t) + P_c(t)\big), \tag{40}$$

where the present value of preferred dividends over a horizon T^* (e.g. 10 years) at discount rate r is

$$\mathrm{PV}_t\big(\textit{preferred dividends}\big) = \Phi_t \cdot \frac{1 - e^{-rT^*}}{r}. \tag{41}$$

Proof. A firm that cannot meet both its hard liabilities (debt) and the present value of its soft liabilities (cumulative preferred dividends) will either (i) default on preferred dividends, triggering governance and market consequences, or (ii) be forced into asset liquidation. The buffer $\varepsilon(P_{nc} + P_c)$ accounts for transaction costs and par-value recovery expectations. □

Definition 4.7 (Extended Survival Probability)**.** *Let the extended distress boundary be*

$$\mathcal{B}_t = D_s(t) + D_c(t) + \mathrm{PV}_t\big(\textit{preferred dividends}\big) + \varepsilon\big(P_{nc}(t) + P_c(t)\big). \tag{42}$$

Define the first-passage time

$$\tau^{\mathrm{ext}} = \inf\big\{u \in [0, T] : A_u \le \mathcal{B}_u\big\}. \tag{43}$$

The extended survival probability is

$$\mathbb{P}\big(\tau^{\mathrm{ext}} > T\big) \approx \frac{\#\{\textit{simulated paths with no hit before } T\}}{\#\{\textit{total paths}\}}. \tag{44}$$

The extended survival probability is strictly lower than the base-model survival probability for any positive preferred dividend obligation, reflecting the additional drag of perpetual fixed-income commitments on BTC-leveraged assets.

4.3 Conversion Optionality

STRK preferred stock carries an embedded conversion right: each share may be converted into $\phi = 0.1$ shares of MSTR common at the holder's election. This creates interplay between the preferred and equity layers.

Definition 4.8 (Conversion Terms)**.** *Let $n_c(t)$ denote the number of outstanding convertible preferred shares (STRK) at time t, with conversion ratio ϕ (shares of MSTR common per preferred share). The* implied conversion strike price *is*

$$K_c = \frac{F_c}{\phi}, \tag{45}$$

where F_c is the liquidation preference (par value) per preferred share.

Proposition 4.9 (Conversion as Embedded American Option)**.** *Each STRK share embeds an American-style call option on MSTR common equity. The holder converts when the equity value received exceeds the present value of retaining the preferred position:*

$$\textit{Convert at } t \quad \Longleftrightarrow \quad \phi \cdot P_t > F_c + \frac{\delta_c F_c}{r_c}, \tag{46}$$

where r_c is the holder's required yield on the preferred, and the right-hand side represents the par value plus the present value of the perpetual dividend stream foregone upon conversion. In

a simplified "intrinsic value" framework:

$$\textit{Convert when } P_t > K_c = \frac{F_c}{\phi}. \tag{47}$$

Remark 4.10. *In practice, because the preferred dividend provides ongoing income, rational holders delay conversion beyond the intrinsic-value trigger. The optimal conversion boundary $P^*(t)$ solves a free-boundary problem analogous to the American call with continuous dividends. For simulation purposes, we use the simplified threshold* (47) *with an optional "conversion premium" multiplier $\psi \geq 1$:*

$$P_t > \psi \cdot K_c. \tag{48}$$

Definition 4.11 (Post-Conversion Dynamics)**.** *When a fraction $\xi_t \in [0, 1]$ of convertible preferred shares convert at time t:*

$$N_t^{\text{sh}} \longrightarrow N_t^{\text{sh}} + \xi_t \, n_c(t) \, \phi, \tag{49}$$

$$n_c(t) \longrightarrow (1 - \xi_t) \, n_c(t), \tag{50}$$

$$\Phi_t \longrightarrow \Phi_t - \xi_t \, \delta_c \, F_c \, n_c(t). \tag{51}$$

Conversion simultaneously dilutes *common shares* (49)*,* reduces *the preferred overhang* (50)*, and* lowers *dividend obligations* (51)*.*

Definition 4.12 (Conversion-Adjusted NAV)**.** *After accounting for potential conversion of all in-the-money convertible preferred shares, the* Conversion-Adjusted NAV *is*

$$\text{NAV}_t^{\text{CA}} = \frac{A_t - D_s(t) - D_c(t) - P_{nc}(t) - \big(n_c(t) - n_c^{\text{ITM}}(t)\big) F_c}{N_t^{\text{sh}} + n_c^{\text{ITM}}(t) \, \phi}, \tag{52}$$

where $n_c^{\text{ITM}}(t)$ is the number of convertible preferred shares for which $\phi \, P_t > F_c$ (i.e. in-the-money). The numerator deducts the par claim of out-of-the-money convertible preferred $\big(n_c - n_c^{\text{ITM}}\big) F_c$, since those shares retain their liquidation preference rather than converting.

Definition 4.13 (Conversion-Adjusted BTC per Share)**.** *The* Conversion-Adjusted BTC per Share *accounts for the maximum dilution from all in-the-money conversions:*

$$B_t^{\text{CA}} = \frac{H_t}{N_t^{\text{sh}} + n_c^{\text{ITM}}(t) \, \phi}. \tag{53}$$

This provides a conservative (fully diluted) measure of BTC ownership per common share.

4.4 Multi-Channel Financing Flywheel

The base model captures BTC accumulation via the holding dynamics $dH_t = \alpha \, \pi_t^+ \, dt + dH_t^{\text{jump}}$. We extend this to disaggregate the financing channels.

Definition 4.14 (Multi-Channel Holding Dynamics)**.** *BTC accumulation is decomposed by financing source:*

$$dH_t = dH_t^{\text{ATM}} + dH_t^{\text{conv}} + dH_t^{\text{pfd}} + dH_t^{\text{ops}}, \tag{54}$$

where:

1. dH_t^{ATM}*: BTC acquired from at-the-market common equity issuance. Proceeds =* (*shares issued*)$\times$ P_t*; BTC purchased = proceeds*$/S_t$*.*

2. dH_t^{conv}*: BTC acquired from convertible debt issuance. Proceeds =* $D_c^{\text{new}}(t)$*; BTC purchased =* $D_c^{\text{new}}(t)/S_t$*.*

3. dH_t^{pfd}*: BTC acquired from preferred stock issuance. Proceeds =* $P^{\text{new}}(t)$*; BTC purchased =* $P^{\text{new}}(t)/S_t$*.*

4. dH_t^{ops}*: BTC acquired from operating cash flows (typically small).*

Each channel entails different cost of capital, dilution characteristics, and market conditions under which it is accessible.

Proposition 4.15 (Channel Selection)**.** *The optimal financing channel depends on the premium* π_t *and market conditions. We model channel intensities as functions of state:*

$$\lambda_t^{\text{ATM}} = \lambda_0^{\text{ATM}}\, \mathbf{1}\{\pi_t > \pi_{\min}^{\text{ATM}}\}\, g_{\text{ATM}}(\pi_t), \tag{55}$$

$$\lambda_t^{\text{conv}} = \lambda_0^{\text{conv}}\, \mathbf{1}\{\sigma_S(t) > \bar{\sigma}\}\, g_{\text{conv}}(\pi_t, r_t), \tag{56}$$

$$\lambda_t^{\text{pfd}} = \lambda_0^{\text{pfd}}\, g_{\text{pfd}}(\pi_t, \text{DCR}_t), \tag{57}$$

where $g_\cdot(\cdot)$ *are non-negative scaling functions and the indicators encode minimum thresholds: ATM issuance requires a positive premium, convertible issuance benefits from high BTC volatility (making the embedded option more valuable), and preferred issuance is viable when dividend coverage is adequate.*

Definition 4.16 (Weighted Average Cost of BTC Acquisition (WACBA))**.** *The* Weighted Average Cost of BTC Acquisition *over the period* $[0, T]$ *accounts for the blended cost of capital across all financing channels:*

$$\text{WACBA}_{[0,T]} = \frac{\sum_{j\in\mathcal{C}} c_j \displaystyle\int_0^T S_u\, dH_u^{(j)}}{\displaystyle\int_0^T dH_u}, \tag{58}$$

where $\mathcal{C} = \{\text{ATM, conv, pfd, ops}\}$ *indexes the channels and* c_j *is the channel-specific cost multiplier that captures dilution and ongoing obligations:*

$$c_{\text{ATM}} = 1 + \frac{\Delta N_{\text{ATM}}^{\text{sh}}}{N_t^{\text{sh}}} \cdot \frac{H_t\, S_t}{\textit{ATM proceeds}}, \tag{59}$$

$$c_{\text{conv}} = 1 + \frac{\textit{coupon PV} + \textit{conversion dilution PV}}{\textit{convertible proceeds}}, \tag{60}$$

$$c_{\text{pfd}} = 1 + \frac{\text{PV}(\textit{preferred dividends})}{\textit{preferred proceeds}}, \tag{61}$$

$$c_{\text{ops}} = 1. \tag{62}$$

Remark 4.17. *The WACBA generalizes the "average BTC purchase price" reported by Strategy. Channels with lower c_j are more efficient per BTC acquired. ATM issuance at high premiums can have $c_{\mathrm{ATM}} < 1$ in BTC-per-share terms (accretive issuance), while preferred issuance carries a perpetual drag via c_{pfd}.*

4.5 Dilution-Adjusted Dynamics

Each financing channel affects the share count differently. We generalize the share-count process from the base model:

$$dN_t^{\mathrm{sh}} = dN_t^{\mathrm{ATM}} + dN_t^{\mathrm{conv_debt}} + dN_t^{\mathrm{conv_pfd}}, \tag{63}$$

where:

- $dN_t^{\mathrm{ATM}} = (\text{ATM proceeds at } t)/P_t$: shares issued via at-the-market program.
- $dN_t^{\mathrm{conv_debt}}$: shares created upon conversion of convertible bonds.
- $dN_t^{\mathrm{conv_pfd}} = \xi_t\, n_c(t)\, \phi$: shares created upon conversion of convertible preferred (STRK).

Proposition 4.18 (BTC-per-Share Accretion Condition)**.** *ATM issuance at time t is BTC-per-share* accretive *if and only if the implied BTC yield per dollar raised exceeds the current BTC per share loading:*

$$\frac{1}{S_t} > \frac{B_t}{P_t} \quad \Longleftrightarrow \quad P_t > B_t \cdot S_t \quad \Longleftrightarrow \quad \frac{E_t}{A_t} > 1 \quad \Longleftrightarrow \quad \pi_t > \log\bigl(\mathrm{ILE}_t\bigr). \tag{64}$$

That is, ATM issuance is accretive when the log-premium exceeds $\log(\mathrm{ILE}_t) = \log\bigl(A_t/(A_t - L_t)\bigr)$. *When $L_t = 0$ this reduces to $\pi_t > 0$; when $L_t > 0$ the threshold is strictly positive.*

Proof. ATM issuance of ΔN shares at price P_t raises $P_t\,\Delta N$ dollars, purchasing $P_t\,\Delta N/S_t$ BTC. Post-issuance BTC per share is

$$B_t^{\mathrm{post}} = \frac{H_t + P_t\,\Delta N/S_t}{N_t^{\mathrm{sh}} + \Delta N}.$$

$B_t^{\mathrm{post}} > B_t$ iff $P_t/S_t > B_t = H_t/N_t^{\mathrm{sh}}$, i.e. $E_t = P_t N_t^{\mathrm{sh}} > H_t S_t = A_t$. Since $E_t = e^{\pi_t}(A_t - L_t)$, this is equivalent to $e^{\pi_t} > A_t/(A_t - L_t) = \mathrm{ILE}_t$, i.e. $\pi_t > \log(\mathrm{ILE}_t)$. □

4.6 Extended Indicators

We define five new indicators that complement the base-model suite (IBGR, ILE, TEE, PMRI, IFRD).

4.6.1 Capital Stack Leverage Ratio (CSLR)

Definition 4.19 (CSLR)**.** *The* Capital Stack Leverage Ratio *measures total claims senior to common equity relative to equity market value:*

$$\mathrm{CSLR}_t = \frac{D_s(t) + D_c(t) + P_{nc}(t) + P_c(t)}{E_t} = \frac{L_t}{E_t}. \tag{65}$$

Unlike the base-model ILE, which captures *elasticity*, the CSLR is a static leverage ratio that directly measures the weight of the liability stack. As preferred stock issuance grows, CSLR rises even if debt remains constant, capturing the layering of soft liabilities.

4.6.2 Preferred Coverage Ratio (PCR)

Definition 4.20 (PCR)**.** *The* Preferred Coverage Ratio *measures the firm's capacity to meet preferred dividend obligations:*

$$\mathrm{PCR}_t = \frac{\mathbb{E}_t^{\mathbb{P}}\big[\Delta A_{t,T}^{+}\big] + \mathrm{CF}_t^{\mathrm{ops}} \cdot T}{\Phi_t \cdot T}, \tag{66}$$

where $\Delta A_{t,T}^{+} = \max(A_T - A_t,\, 0)$ *is expected BTC appreciation over horizon* T*, and* Φ_t *is the annual preferred dividend obligation from* (38)*.*

$\mathrm{PCR}_t > 1$ indicates that expected asset appreciation covers preferred dividends; $\mathrm{PCR}_t < 1$ signals potential dividend stress and increased probability of governance consequences (e.g. board representation rights for preferred holders).

4.6.3 Dilution-Adjusted IBGR

Definition 4.21 (Dilution-Adjusted IBGR)**.** *The* Dilution-Adjusted IBGR *measures BTC growth per share net of all dilution sources, including potential preferred stock conversions:*

$$\mathrm{IBGR}_t^{\mathrm{DA}} = \frac{1}{B_t^{\mathrm{CA}}} \mathbb{E}_t^{\mathbb{P}} \left[\frac{B_{t+\Delta t}^{\mathrm{CA}} - B_t^{\mathrm{CA}}}{\Delta t} \right], \tag{67}$$

where B_t^{CA} *is the Conversion-Adjusted BTC per Share from* (53)*. This differs from the base-model per-share IBGR by using the fully diluted share count.*

In practice, $\mathrm{IBGR}_t^{\mathrm{DA}}$ is estimated via Monte Carlo by simulating joint paths of $(S_t, H_t, N_t^{\mathrm{sh}}, n_c(t))$ under $\mathbb{P}$ and computing sample means of the ratio.

4.6.4 Effective Leverage Elasticity (ELE)

Definition 4.22 (ELE)**.** *The* Effective Leverage Elasticity *generalizes ILE to account for the preferred stock buffer layer between debt and common equity:*

$$\mathrm{ELE}_t = \left.\frac{\partial \log E_t}{\partial \log S_t}\right|_{\pi_t \text{ fixed}} = \frac{H_t\, S_t}{H_t\, S_t - D_s(t) - D_c(t) - P_{nc}(t) - P_c(t)} = \frac{A_t}{A_t - L_t}, \tag{68}$$

whenever $A_t > L_t$*.*

Proposition 4.23 (Ordering of Elasticities)**.** *When preferred stock outstanding is positive,*

$$\mathrm{ELE}_t \;\geq\; \mathrm{ILE}_t = \frac{A_t}{A_t - D_s(t) - D_c(t)}, \tag{69}$$

with equality iff $P_{nc}(t) + P_c(t) = 0$. The incremental leverage from the preferred layer is

$$\Delta\beta_t^{\text{pfd}} = \text{ELE}_t - \text{ILE}_t = \frac{A_t\big(P_{nc}(t) + P_c(t)\big)}{\big(A_t - D_t\big)\big(A_t - L_t\big)}, \tag{70}$$

where $D_t = D_s(t) + D_c(t)$.

Proof. Both ELE_t and ILE_t have the form $A_t/(A_t - X)$ with $X_{\text{ELE}} = L_t \geq X_{\text{ILE}} = D_t$. The function $f(X) = A_t/(A_t - X)$ is increasing in X for $X < A_t$, yielding the inequality. The difference follows by algebra. □

The extended Total Equity Elasticity becomes

$$\beta_{\text{TOT}}^{\text{ext}}(t) = \gamma_{\pi S} + \text{ELE}_t, \tag{71}$$

replacing ILE_t with ELE_t in the base formula.

4.6.5 Capital Efficiency Ratio (CER)

Definition 4.24 (CER)**.** *The* Capital Efficiency Ratio *measures the BTC-per-share accretion generated per unit of capital stack leverage:*

$$\text{CER}_t = \frac{\text{IBGR}_t^{\text{DA}}}{\text{CSLR}_t}. \tag{72}$$

High CER indicates that additional leverage is being deployed productively (generating BTC accretion per share), while low CER signals that the capital stack is growing faster than per-share BTC accumulation — a sign of diminishing returns to the flywheel strategy.

4.7 Structural Relationships and Summary

Theorem 4.25 (Flywheel Decomposition)**.** *Under the extended model, the instantaneous change in common equity value decomposes as:*

$$\frac{dE_t}{E_t} = \underbrace{\text{ELE}_t \cdot \frac{dS_t}{S_t}}_{\textit{leveraged BTC return}} + \underbrace{d\pi_t}_{\textit{premium change}} - \underbrace{\frac{\Phi_t\, dt}{A_t - L_t}}_{\textit{preferred dividend drag}} + \underbrace{\frac{S_t\, dH_t}{A_t - L_t}}_{\textit{BTC accretion lift}} + O(dt). \tag{73}$$

Proof. From $E_t = e^{\pi_t}(A_t - L_t)^+$ with $A_t = H_t S_t$, apply Itô's lemma to $\log E_t = \pi_t + \log(A_t - L_t)$ and note that preferred dividends reduce A_t at rate Φ_t (dividends are paid from assets), while BTC acquisitions increase A_t at rate $S_t\, dH_t$. Collecting terms yields the decomposition. □

This decomposition makes explicit the tension at the heart of the Strategy model:

- The BTC accretion lift is powered by the financing flywheel (Section 4.4).
- The preferred dividend drag is the cost of funding via preferred equity, which grows monotonically with issuance.
- The premium change can amplify or dampen returns depending on market sentiment.

- Effective leverage ELE_t increases with both debt and preferred issuance, amplifying BTC returns in both directions.

Table 1 summarizes the complete indicator suite.

Table 1: Summary of structural indicators: base model and extensions.

Indicator	**Type**	**Description**
ILE	Base	Structural leverage elasticity (debt only)
TEE	Base	Total equity elasticity incl. premium response
PMRI	Base	Premium z-score under OU stationarity
IBGR	Base	Expected BTC growth rate (total and per-share)
IFRD	Base	Funding requirement distribution
CSLR	Extended	Total claims / equity (full capital stack leverage)
PCR	Extended	Expected asset appreciation / preferred dividends
$\mathrm{IBGR}^{\mathrm{DA}}$	Extended	Dilution-adjusted BTC growth per share
ELE	Extended	Leverage elasticity incl. preferred buffer
CER	Extended	BTC accretion per unit leverage

5 Results

We calibrate the model using publicly available data on Strategy (MSTR), Bitcoin, and the company's balance sheet through December 1, 2025. This section reports the estimated parameters, computed indicators, and their economic interpretation. All figures referenced below are produced by the calibration pipeline and are included in the appendix.

5.1 Calibrated Parameters

Table 2 reports the full set of estimated parameters.

Table 2: Calibrated Model Parameters (as of December 1, 2025)

Parameter	**Symbol**	**Value**	**Description**
BTC Price Dynamics			
BTC drift	μ_S	0.140	Annualized historical return
BTC volatility	σ_S	0.488	Annualized historical volatility
Premium OU Process			
Mean-reversion speed	κ	4.081	Annual rate of reversion to θ
Long-run mean	θ	0.705	Corresponds to a $\approx 2.0\times$ NAV premium
Premium volatility	σ_π	1.417	Annualized diffusive vol of log-premium
BTC–premium correlation	ρ	-0.491	Negative: premium compresses on BTC rallies
Premium–BTC sensitivity	$\gamma_{\pi S}$	-1.480	Regression coefficient
Holdings Dynamics			
Continuous drift	α	0.000	Not significantly different from zero
Jump intensity	λ_M	17.295	$\approx$17.3 financing events per year
Mean jump size	$\mathbb{E}[\Delta H]$	6,887	BTC per event
Balance Sheet (Initial Conditions)			
BTC price	S_0	\$86,286	
BTC holdings	H_0	649,870	
Outstanding debt	D_0	\$8.22B	Face value
Shares outstanding	N_0^{sh}	267.7M	
Current log-premium	π_0	-0.042	Slightly below NAV

BTC dynamics. The estimated BTC annualized volatility of 48.8% is consistent with the cryptocurrency's historical behavior over the sample period. The positive drift of 14.0% reflects the secular upward trend in BTC price under the physical measure.

Premium process. The OU parameters reveal fast mean-reversion ($\kappa = 4.08$, implying a half-life of approximately $\ln 2/\kappa \approx 62$ calendar days) toward a long-run log-premium of $\theta = 0.705$, corresponding to equity trading at roughly $2.0\times$ NAV in the long run. The premium volatility is high ($\sigma_\pi = 1.42$), reflecting the large swings in market sentiment around BTC treasury vehicles.

The estimated correlation $\rho = -0.49$ is economically significant and negative: when BTC price rises, the premium tends to compress, and vice versa. This suggests that BTC rallies are partially "priced in" through NAV increases, reducing the sentiment component. The regression coefficient $\gamma_{\pi S} = -1.48$ confirms this quantitatively—a 1% BTC increase is associated with a 1.48 percentage point decrease in the log-premium, on average.

Holdings dynamics. The continuous accumulation drift α is estimated at zero, indicating that all meaningful BTC acquisition occurs through discrete financing events rather than continuous purchasing. The jump intensity $\lambda_M = 17.3$ events per year implies roughly one financing event every three weeks on average. The mean purchase size of 6,887 BTC per event, at late-

2025 prices, corresponds to approximately $594 million per event—consistent with the scale of Strategy's ATM and convertible offerings.

5.2 Computed Indicators

Table 3 summarizes the five diagnostic indicators.

Table 3: Model-Implied Indicators (as of December 1, 2025)

Indicator	**Value**	**Interpretation**
IBGR (total)	18.3%	Annualized BTC accumulation rate
IBGR (per-share, 3yr)	18.4%	Annualized BTC-per-share growth
ILE	1.172	Structural leverage elasticity
TEE	−0.308	Total equity–BTC elasticity
PMRI	−1.506	Std. devs below long-run mean premium
IFRD (mean, 3yr)	$38.3B	Expected cumulative funding need
IFRD (95th pctl, 3yr)	$80.3B	Tail funding need
Survival prob (3yr, $\varepsilon = 0$)	99.42%	No-distress probability
Survival prob (3yr, $\varepsilon = 10\%$)	99.32%	With 10% buffer

5.2.1 IBGR: Implied BTC Growth Rate

The model implies a total BTC accumulation rate of 18.3% annually, driven entirely by discrete financing events (since $\alpha = 0$). Strikingly, the per-share IBGR of 18.4% is nearly identical, indicating that share dilution from equity issuance has been almost perfectly offset by the volume of BTC acquired—the flywheel has been *highly accretive* on a per-share basis.

This finding has direct implications for valuation: if the market expects the flywheel to continue operating at this pace, the premium over NAV reflects an expectation of compounding BTC-per-share growth at approximately 18% annually, in addition to any appreciation in BTC itself.

5.2.2 ILE: Implied Leverage Elasticity

The ILE of 1.172 indicates moderate structural leverage. Strategy's BTC assets of approximately $56.1 billion exceed its $8.2 billion debt by a factor of 6.8, producing a leverage elasticity only modestly above 1.0. This means that, holding the premium fixed, a 10% BTC decline would produce approximately an 11.7% equity decline—manageable leverage that reflects the large cushion between assets and debt.

Figure 2 (see results/figures/ile_tee_timeseries.png) shows the time series of ILE, which has declined from significantly higher levels in 2021–2022 when BTC prices were lower relative to debt, to its current low level as BTC appreciation has dwarfed the debt stack.

5.2.3 TEE: Total Equity Elasticity

The TEE of -0.308 presents a striking contrast to the ILE. While the balance-sheet leverage contributes positively ($+1.172$), the premium's negative sensitivity to BTC ($\gamma_{\pi S} = -1.48$) more than offsets it. The net effect is that, empirically, MSTR equity has moved *inversely* to BTC on average in recent data—a counterintuitive result driven by the premium dynamics.

This decomposition is the TEE's primary analytical value: it separates the "structural" BTC exposure (ILE, which is always ≥ 1) from the "sentiment" exposure ($\gamma_{\pi S}$, which can be any sign), revealing that short-term MSTR returns are dominated by premium fluctuations rather than BTC fundamentals.

5.2.4 PMRI: Premium Mean-Reversion Index

The PMRI of -1.51 indicates that the current premium ($\pi_0 = -0.042$, i.e., equity trading approximately at NAV) is 1.5 stationary standard deviations below the long-run mean ($\theta = 0.705$, i.e., $\approx 2\times$ NAV). Under the OU dynamics, this represents a mean-reverting opportunity: the model expects the premium to drift upward toward its long-run level with a half-life of roughly 62 trading days.

Figure 1 (see results/figures/premium_timeseries.png) displays the premium time series, illustrating the mean-reverting behavior and the current below-average level.

5.2.5 IFRD: Implied Funding Requirement

The IFRD indicates that sustaining the model-implied BTC accumulation trajectory over three years requires, in expectation, approximately \$38.3 billion in financing, with a 95th percentile tail of \$80.3 billion. These are large numbers in absolute terms, but Strategy has demonstrated the capacity to raise capital at this scale through its ATM program and convertible issuances (the 21/21 plan alone targeted \$42 billion in financing).

Figure 3 (see results/figures/ifrd_histogram.png) shows the simulated IFRD distribution, which is right-skewed, reflecting the positive correlation between BTC price and funding costs (higher BTC prices make each unit of BTC more expensive to acquire).

5.2.6 Survival Probability

The three-year survival probability of 99.42% (99.32% with a 10% buffer) indicates that the current capital structure is highly resilient to distress. The large gap between BTC asset value (\$56.1B) and debt (\$8.2B) means that BTC would need to decline by approximately 85% before assets breach the debt boundary—an extreme but not unprecedented scenario for cryptocurrency. The survival analysis incorporates the full joint dynamics, including the possibility of further BTC accumulation via the flywheel, which acts as a partial buffer (acquiring BTC at lower prices when the premium permits).

Figure 5 (see results/figures/assets_debt_nav.png) illustrates the historical evolution of assets, debt, and NAV, showing the growing cushion over time.

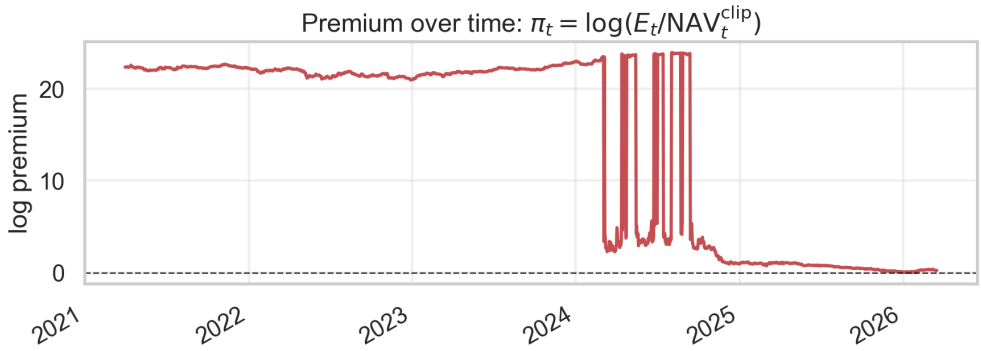

Figure 1: Historical log-premium $\pi_t = \log(E_t/\mathrm{NAV}_t^{\mathrm{clip}})$. The dashed line indicates the OU long-run mean $\theta = 0.705$. The premium exhibits clear mean-reverting behavior around this level.

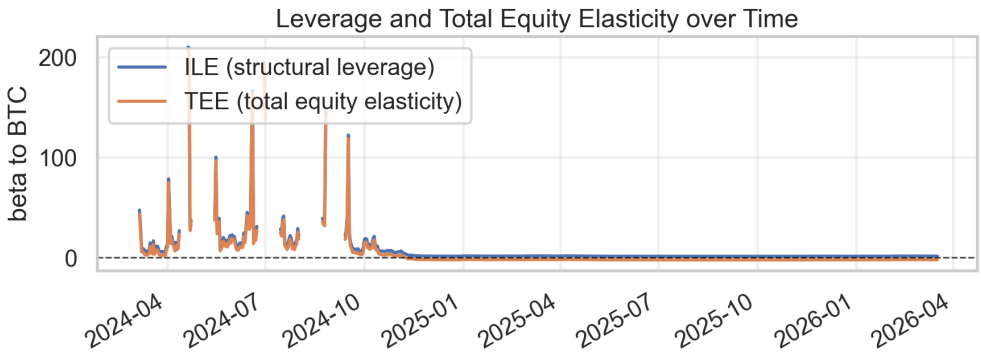

Figure 2: Time series of Implied Leverage Elasticity (ILE) and Total Equity Elasticity (TEE). ILE has declined as BTC appreciation outpaced debt growth. TEE reflects the net of ILE and the negative premium–BTC sensitivity.

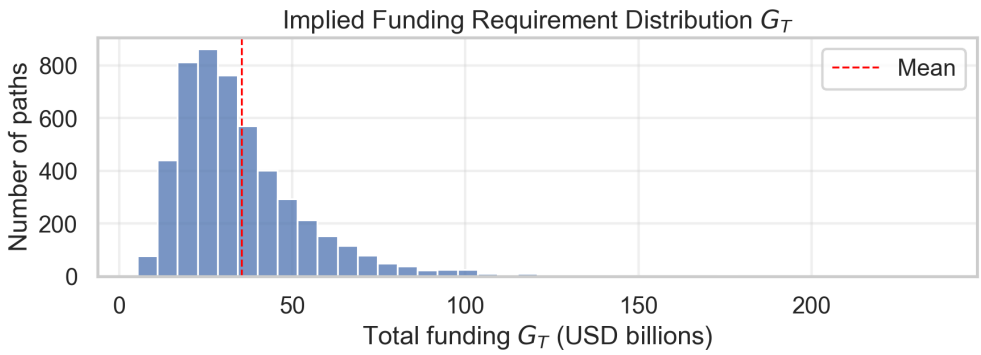

Figure 3: Simulated three-year Implied Funding Requirement Distribution (IFRD). The distribution is right-skewed, with mean \$38.3B and 95th percentile \$80.3B.

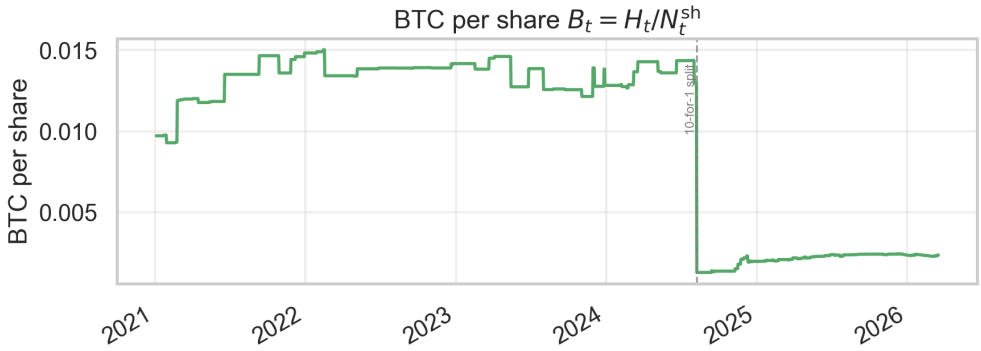

Figure 4: BTC per share ($B_t = H_t/N_t^{\text{sh}}$) over time. The upward trend confirms that Strategy's flywheel has been accretive despite significant share dilution.

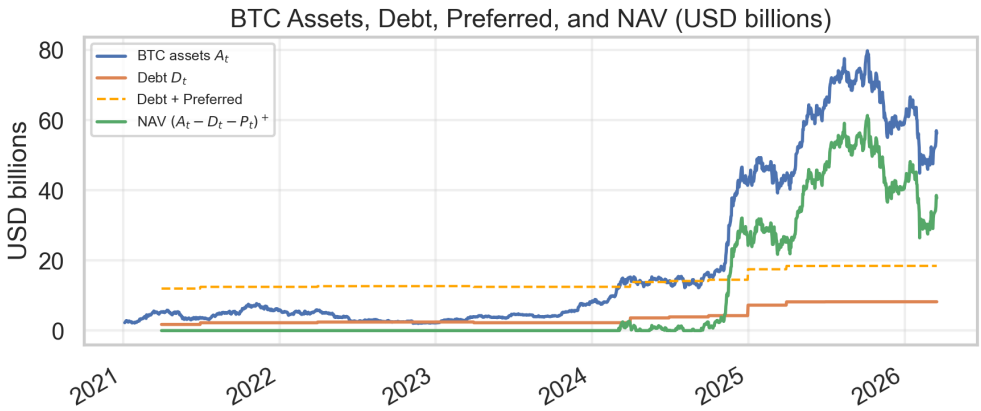

Figure 5: BTC asset value ($A_t = H_t S_t$), outstanding debt (D_t), and NAV ($(A_t - D_t)^+$). The growing gap between assets and debt underpins the high survival probability.

6 Discussion

6.1 What the Model Tells Us About Strategy's Viability

The calibrated model paints a picture of a company that has, through late 2025, successfully executed its leverage flywheel. Three findings stand out:

First, the flywheel has been accretive. The near-equality of total IBGR (18.3%) and per-share IBGR (18.4%) demonstrates that Strategy has acquired BTC at prices and share-issuance ratios that deliver increasing BTC per share. This is the core value proposition: common shareholders receive compounding exposure to BTC without directly purchasing it. The pure jump model ($\alpha = 0$) confirms that accretion occurs through discrete, large financing events rather than gradual accumulation—consistent with the company's pattern of multi-billion dollar ATM programs and convertible offerings.

Second, balance-sheet leverage is currently modest. An ILE of 1.17 means the equity's structural exposure to BTC barely exceeds 1:1. The 2020–2022 period saw much higher

leverage (when BTC prices were lower relative to debt), but BTC appreciation has dramatically delevered the balance sheet. The \$8.2 billion debt stack is a small fraction of the \$56.1 billion BTC portfolio, providing substantial cushion. The 99.4% three-year survival probability reflects this buffer.

Third, the premium is currently depressed. The PMRI of -1.51 suggests the market is pricing Strategy conservatively relative to the model's long-run equilibrium. Under the calibrated OU dynamics (half-life of $\sim$62 calendar days), the premium is expected to mean-revert upward. Whether this represents a buying opportunity depends on whether the OU calibration (fitted primarily to the 2020–2025 sample) remains valid as the capital structure evolves with new preferred layers.

6.2 Implications for Other BTC Treasury Companies

Strategy's success has spawned imitators: several companies have adopted or announced Bitcoin treasury strategies, and the analytical framework developed here applies directly. The model's key inputs—BTC holdings, debt schedule, share count, and premium—are publicly observable for any such company, enabling cross-sectional comparison of flywheel efficiency and structural health.

The indicators have natural comparative interpretations. A company with high ILE and high IBGR is aggressively but accretively leveraging BTC. A company with high ILE but low or negative per-share IBGR is diluting shareholders without sufficient BTC accumulation. The PMRI can identify relative value across BTC treasury vehicles by comparing each company's premium deviation from its own long-run mean.

6.3 Model Limitations

Several limitations warrant discussion:

Stationarity of the OU premium. The OU specification assumes a constant long-run mean θ and mean-reversion speed κ. In practice, θ may shift as the BTC treasury model becomes better understood (or as new competitors enter), and κ may vary with market microstructure and investor base composition. A regime-switching extension would address this but at the cost of additional parameters.

Exogenous debt. Modeling debt as deterministic ignores the strategic timing of issuance and the option value of convertible bonds. In reality, management times debt issuance opportunistically (issuing when premia are high and credit markets receptive), creating endogeneity that our model does not capture. A fuller model would make debt issuance intensity depend on the premium and BTC price, mirroring the holding dynamics.

Risk-neutral vs. physical measure. All indicators are computed under the physical measure, appropriate for real-world risk assessment. However, for pricing derivatives on MSTR equity or for risk-neutral valuation of the convertible bonds, a measure change would be needed, requiring specification of market prices of risk for both BTC and premium factors.

Sample period. The calibration uses data from 2020–2025, a period of generally favorable conditions for both BTC and the treasury model. Parameters estimated from this sample

(particularly μ_S and θ) may overstate the attractiveness of the strategy in a more adverse macro environment.

Correlation stability. The negative ρ and $\gamma_{\pi S}$ are estimated as constants, but the relationship between BTC returns and the premium may be state-dependent—for instance, the premium may become more sensitive to BTC in high-leverage states or during market stress.

6.4 Preferred Stock Extensions

Strategy's capital stack has expanded significantly with the introduction of perpetual preferred stock series. Extending the model to accommodate these layers requires:

Priority waterfall. In liquidation and for dividend distribution, the capital stack follows: senior debt → convertible bonds → preferred stock (by seniority) → common equity. NAV must be allocated through this waterfall, with each tranche receiving its claim before the residual passes to the next.

Perpetual dividend obligations. Unlike debt maturities, preferred stock carries perpetual fixed dividends (8% for STRK, 10% for STRF). These create ongoing cash flow demands that affect the survival analysis—distress is no longer solely an asset-vs-debt question but includes the ability to service dividend obligations from BTC sales or new issuance.

Conversion optionality. STRK's convertibility into MSTR common stock at a fixed ratio (0.1 shares per STRK share) creates an embedded option that affects both the preferred valuation and the share count dynamics. When MSTR trades above the implied conversion price, STRK holders have an incentive to convert, diluting common equity but reducing preferred dividend obligations.

Incorporating these features into the simulation engine is a natural next step. The framework developed here—with its joint evolution of BTC price, premium, holdings, and share count—provides the scaffolding; what remains is to layer the priority waterfall on top of the existing NAV computation and add dividend cash-flow accounting to the survival analysis.

6.5 Broader Implications for Corporate Finance

The Bitcoin treasury vehicle represents a novel corporate form that challenges traditional frameworks. It is not a holding company (it actively trades), not a fund (it uses leverage and issues preferred equity), and not a traditional operating company (its "operations" are financial engineering). The structural model developed here suggests that this form can be viable—with 99%+ survival probabilities—when operated with discipline (moderate leverage, accretive issuance). The key risk is not BTC volatility per se, but the potential for the flywheel to become *dilutive*: if management issues shares or converts at unfavorable ratios, per-share BTC declines, the premium collapses, and the flywheel reverses. Our indicators—particularly the per-share IBGR and the PMRI—are designed to detect this transition early.

7 Conclusion

This paper develops a stochastic structural framework for valuing Bitcoin treasury vehicles—publicly traded companies whose primary asset is BTC, acquired through a leverage flywheel

of equity, debt, and preferred stock issuance. We jointly model BTC price dynamics, the equity-over-NAV premium as an Ornstein–Uhlenbeck process, BTC holding accumulation as a compound Poisson process linked to the flywheel, and share-count evolution from issuance and conversion.

From this framework we derive five diagnostic indicators: the Implied BTC Growth Rate (IBGR), Implied Leverage Elasticity (ILE), Total Equity Elasticity (TEE), Premium Mean-Reversion Index (PMRI), and Implied Funding Requirement Distribution (IFRD). Together, these indicators decompose the valuation of a BTC treasury vehicle into its structural components—balance-sheet leverage, sentiment premium, accretion efficiency, and financing sustainability—providing investors and analysts with a rigorous toolkit for assessing these novel entities.

Calibrating to Strategy (MSTR) through December 2025, we find that the flywheel has been highly accretive (18.3% annualized BTC growth, nearly identical per-share), balance-sheet leverage is currently moderate (ILE = 1.17), and the capital structure supports a three-year survival probability exceeding 99.4%. The premium currently sits 1.5 standard deviations below its long-run mean, suggesting potential for mean-reversion.

These results are conditioned on the calibration sample and the model's assumptions of parameter stationarity. As Strategy's capital structure continues to evolve—with multiple preferred stock series, expanding ATM programs, and potential international vehicles—the framework presented here provides a foundation for increasingly granular analysis. The extension to multi-layer preferred capital stacks, outlined in the discussion, is a natural and important direction for future work.

References

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